# MITIGATING COLOUR BIAS BY "UN-LEARNING" COLOUR INFORMATION IN THE CONTEXT OF SEMANTIC SEGMENTATION

**MSc Data Science Thesis – Interim Report**

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# 1 Introduction

Recent years have seen a surge in the development of artificial intelligence (AI) systems; largely fuelled by the symbiosis of deep learning progress, and advancements in micro/nanochip manufacturing – harnessing remarkable compute power. Ubiquitous deployment of AI throughout modern society means that practitioners have an ethical and moral responsibility in the dissemination of this cutting-edge technology.

Within the field of deep learning, convolutional neural networks (CNNs) have gained substantial traction in the last decade, and their performance on computer vision tasks is state-of-the-art [1] [2] [3]. Due to the complex nature of these systems, a 'black box' stigma is often attached to them; since deep learning networks are now achieving unprecedented levels of accuracy, external scrutiny and media spotlight demands a push towards transparency, fairness, and accountability. This has led to the more recent movement of 'explainable artificial intelligence' (XAI). To be clear, neural networks can be interrogated for interpretability [4] [5] –yielding detailed reverse engineered optimised feature maps, and heatmaps denoting important regions that the model has based its decision on. These solutions tackle the *what* and *where* of the common issues related to explainability, however, we currently can't ask the network *why* it made such a decision.

The crux of developing fair AI is the elimination of bias –a long sought issue in any statistical modelling application. Bias can manifest itself in a multitude of guises which are generally quite context specific. For the purposes of this study we will focus on algorithmic bias. Bias of this nature can creep into our models from training data, meaning our models exhibit the same systemic discrimination found in the wild. This bias can often be unknown and undetected – making for a particularly insidious force. Indeed, George Santayana warned us that *"those who do not remember the past are condemned to repeat it"*. If bias is allowed to creep into our models undetected, we risk propagating this bias throughout society; eventually distilling into a self-fulfilling prophecy —one which automates inequality.

Natural language processing is particularly susceptible to this prejudice. Training data comprises of swathes of online corpora, and the sheer scale of data means inspecting the training data manually for bias is completely impractical. Sheng et al. [6] demonstrate quite a shocking example of societal stereotyping using a next sentence generation model. Recently, in computer vision fields, it has been shown [7] that software deployed in facial recognition systems perform better on white skinned male individuals rather than other minority races, particularly females. This racial and gender discrimination is caused by biased training data comprising of under-represented minority classes. Often the open-source datasets which facial recognition models are trained on come from celebrity faces [8] [9] from the western world or public figures mined from internet image searches. Such datasets are not representative of the whole population; more recent examples of balanced datasets [10] have been compiled to evaluate societal bias in landmark facial recognition APIs.

More generally, CNNs have been shown to be biased [11] towards texture (rather than say, shape, colour etc.). Interestingly, the CNNs adopt an Occam's Razor philosophy in their learning process; indeed, if the texture information is sufficient to perform image categorisation well, then why opt for another cue? This becomes problematic when the CNNs pick up on the wrong cues, especially the bias within the data and use that to make decisions. Furthermore, we cannot easily push the model to use the features we want.

(a) Urban Scene Example (b) Semantic Segmentation (c) Instance Segmentation (d) Panoptic Segmentation

Figure 1: Types of image segmentation: **(b)** Pixel-wise categorisation of different semantic classes, e.g. different cars belong to the 'car' class and different people to the 'people' class. **(c)** Each instance of a *thing* has a different label. So each person is categorised separately, background *'stuff'* classes are ignored. **(d)** A hybrid of (b) and (c). This project will focus on Semantic Segmentation. Exemplar images taken from [12].

This research focusses on colour bias which exists within urban scenes. Urban scene segmentation is at the heart of autonomous vehicle (AV) technology, and to date many network architectures have been developed achieving impressive accuracy. This blinkered push towards optimal accuracy often neglects to consider biases within the training data, thus, any advancements in bias mitigation within the realm of image segmentation are novel. Colour bias can manifest in many ways within urban scenes. Successful AV technology must be dependable in a multitude of conditions, including: densely populated urban streets, fog, rain, snow, glare, and seasonal changes. This highly variable distribution of data poses a great challenge. Clearly, the same tree that a segmentation model picks up on in summer might look very different in winter, or in New York would a CNN lean to categorising all yellow boxes as cars due to the number of taxis? We humans are extremely adaptable and can make decisions to correct our actions depending on external stimulus; machines however, struggle to perform well in edge cases or situations not encountered in the training phase. Thus, a push towards robustness and generalisation is paramount for the evolution of safe AV technology.

The umbrella of image segmentation covers three main domains (see Figure 1): instance segmentation, semantic segmentation, and more recently a hybrid of both – panoptic segmentation. In this work we will consider semantic segmentation, which concerns categorising each pixel in an image into one of $n$ predefined classes. This process splits the image into different regions based on what the pixels show and is the adopted methodology for AV systems.

## 2 Aim and Objectives

### 2.1 Aim

This work seeks to increase the robustness of CNNs via the mitigation of algorithmic bias. We aim to implement an 'unlearning' procedure within the network architecture itself rather than increase generalisability via augmentation of the input data. The unlearning procedure uses a multi-headed network to adversarially remove a certain feature using reverse gradient loss [13]. Semantic segmentation is used as a vehicle to assess the effectiveness of such a system, albeit the core principle, should, in theory, be able to be applied to any deep learning architecture.

### 2.2 Objectives

This work will focus on three primary objectives:

- Determine whether using a multi-headed network architecture to adversarially remove the bias from urban scenes in a pixel-wise colour classification model is effective. This method requires knowing what the bias is.
- Compare whether the approach proposed in [14] may be optimal by assessing the performance of using a Variational Autoencoder (VAE) to create a latent representation of the input data and thus representatively sample the data at training time accordingly. The power in this method lies in the VAE determining what the bias actually is. To date, application of this method to semantic segmentation tasks have not been studied, and it isn't understood if a highly variable input image such as urban scenes would benefit from this method.
- The hypothesis of texture and colour acting as bias in the task of semantic segmentation is tested by comparing the results of all methods to the accuracy of a greyscale baseline training set containing no colour information, and thus no bias in our context, and also the natural uncorrupted images.

Optional objectives time permitting:

- Explore the proposed solution using panoptic segmentation [12] networks. Theoretically the solution should be generalisable to other domains
- Formulate and implement a novel idea inspired by Squeeze and Excitation networks [15] essentially re-weighting the feature maps in the first layer of the CNN depending on how much colour information they contain, penalising maps focussing solely on colour. In theory this should push the model to more geometric-based cues.

## 3 Overview of progress

A thorough literature review has taken place. A research net was cast wide initially, covering applications of image segmentation, evolution of autonomous vehicle technology, governance and acceptance of AI technologies, and societal biases. More specifically, seminal deep CNN semantic segmentation architectures have been studied at depth, progressing to a review of techniques addressing robustness and bias mitigation.

Under the umbrella of bias removal, the taxonomy forms three natural groupings:

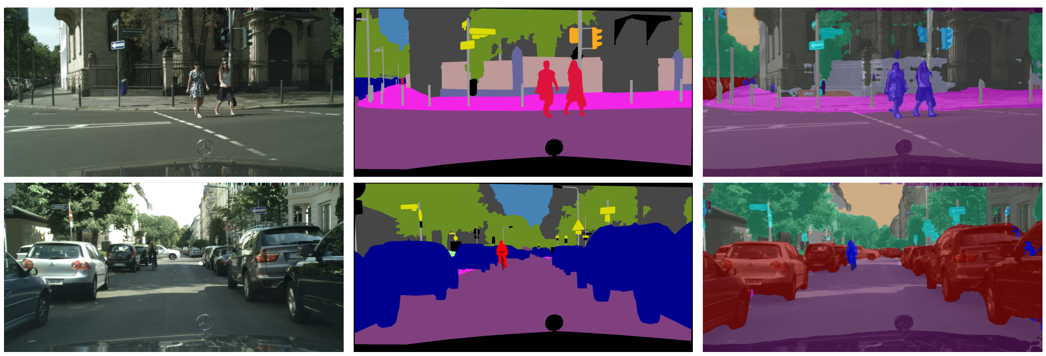
- Those seeking to increase generalisation and thus reduce the effects of bias via image augmentation.
- Those using the network architecture to attempt to remove or mitigate a *known* bias.

- Those attempting to *learn* the bias within a given dataset and mitigate it accordingly.

Most developed techniques to tackle this issue rely on a known bias [13] [16] [17], in the burgeoning world of big data, it is often unsurmountable to assess data for such bias, furthermore we are also at the mercy of our own biased representations. Amini et al. [14] tackle this issue with the use of variational autoencoders (VAE's). Their proposed model actually learns, in an unsupervised manner, the latent structure of the input data and adaptively uses this learned latent distribution to selectively up-sample under-represented data points. Although Amini et al. demonstrate this technique through racial and gender bias in facial recognition systems the idea itself is generalisable to multiple domains. This project aims to implement this schema in the context of semantic segmentation; to see if the concept can handle the highly variable domain of urban scenes.

The main risk to project success is the highly variable domain of urban scenes. Multi-headed models which have been used to remove bias [16] have been published as a proof-of-concept on toy datasets such as MNIST. Thus the performance of putting the theory into practice using edge data will be interesting.

The project leverages Google's Colab Pro [18] web IDE, utilizing the PyTorch [19] library for Python with 12GB RAM and 150GB of cloud storage. Security and confidentiality is not a concern given the public nature of the data used. Indeed, any trained networks we create push towards a philanthropic cause of bias mitigation in AI –meaning public distribution is encouraged.



(a) Raw Images (b) Ground Truth (c) DeepLabv3 Prediction

Figure 2: Implementation progress training a Deeplabv3 segmentation network on the cityscapes dataset.

To date the DeepLabv3 [20] network with a pretrained ResNet18 [21] backbone has been trained for 185 epochs and is producing some initial results, figure 2 shows some qualitative results compared to the ground truth and raw image. Results are impressive after $\approx$ 20 hours training. Figure 3 shows the loss curves after 100 epochs where a steady convergence to optimisation can be seen. Ideally training can be prolonged until we reach the mean intersection over union (mIoU) of 81.3% on the cityscapes test dataset *(achieved by Chen et al. [20])*. A SegNet [22] model is also in an implementation phase but only in the early stages of training.

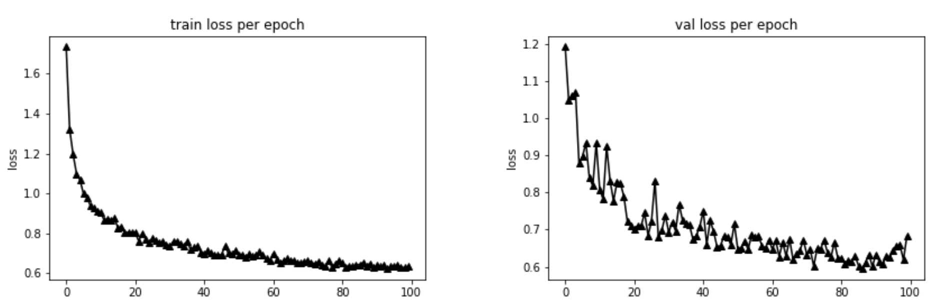



Figure 3: Training and validation loss curves during the first 100 epochs of training showing steady convergence

## 4 Project Plan

The project time frame spans from 26th April 2021 to the 27th August 2021. A detailed visual plan is provided in Figure 4. Preliminary work included research into the field of image segmentation, autonomous vehicles and societal bias. A review of related literature has been compiled ready for the final report. The implementation phase of the project is scheduled to run 18th May to the 10th August. Software will be developed in an agile manner with regular testing and iterations. Results will be evaluated on their ability to successfully unlearn colour bias in the context of semantic segmentation, it is expected that this may come at a sacrifice of overall accuracy score; however a more robust model should have been reached. Time allowing, other datasets can be explored with the proposed architectures, and performance at panoptic segmentation can be assessed. Project deliverables are defined and progress is monitored.

## Project Repository

— www.github.com/JackStelling/BiasMitigation

## References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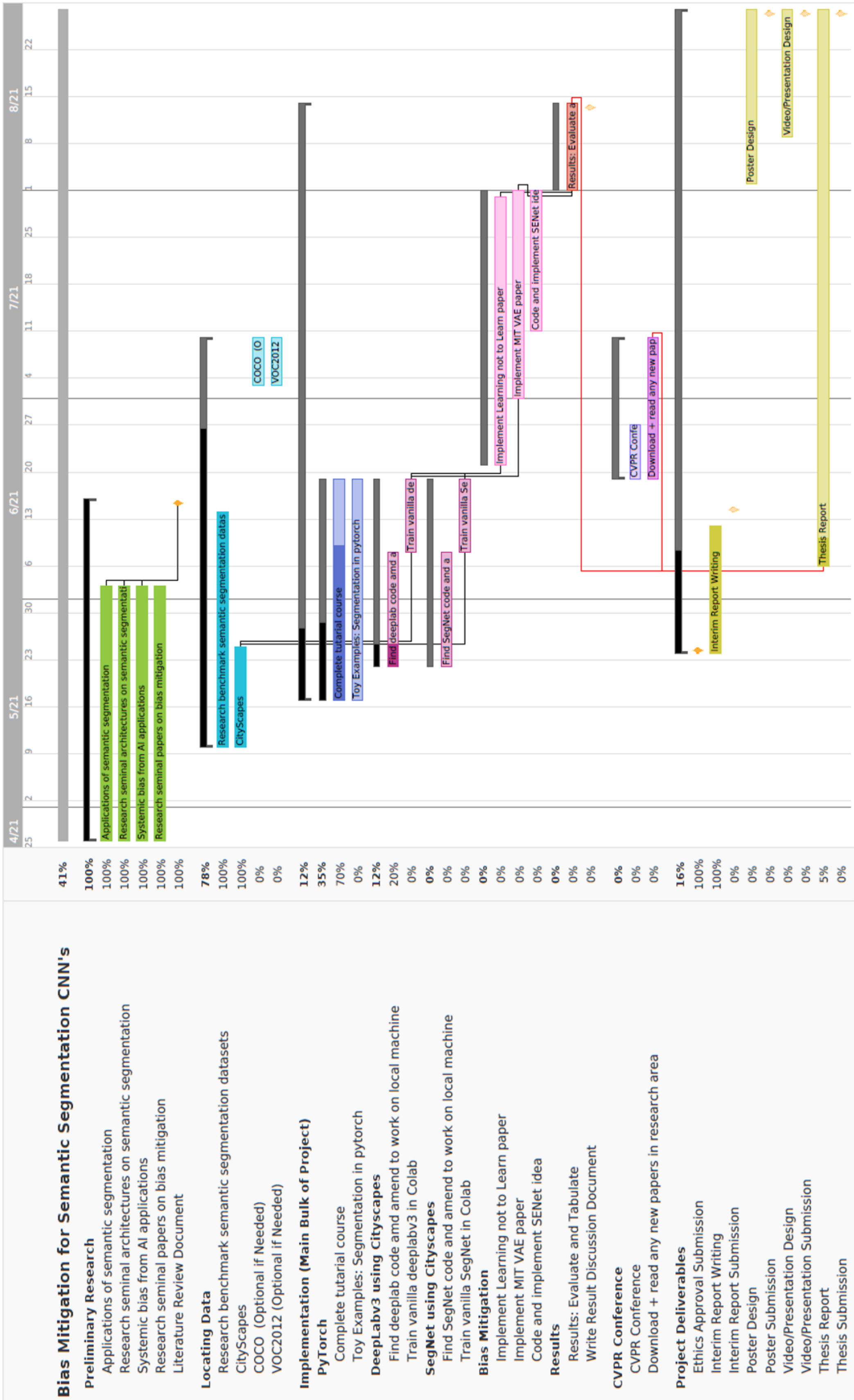


Figure 4: A Gantt Chart of the proposed project. Tasks are to scale as per date range, and dependencies are shown as lines.

# DATA MANAGEMENT PLAN.

| 0. Project title, author, version and date | | |
|---|---|---|
| ***Project:*** *Mitigating Bias in the context of Semantic Segmentation.* | | |
| ***Author:*** *Jack Stelling* | ***Version:*** *1* | ***Date:*** *8th June 2021* |

## 1. Description of the data

### 1.1 Type of study

*Studying the effect of different neural network architectures on the removal of bias from deep convolutional neural networks used for semantic segmentation. Specifically, the datasets which we will be considering are 'dashcam' images of urban scenes used for training autonomous vehicle technology.*

### 1.2 Assessment of existing data

*Only publicly available datasets are considered for this project. The 'Cityscapes' (https://www.cityscapes-dataset.com/) dataset will provide the bulk of data to make initial investigations into the success of any proposed models. Cityscapes is the most popular benchmark dataset for segmentation tasks. If time allows, results can be cross examined with other benchmark collections such as: COCO (https://cocodataset.org/), VOC2012 (http://host.robots.ox.ac.uk/pascal/VOC/voc2012/), KITTI (http://www.cvlibs.net/datasets/kitti/) and offshoots of Cityscapes with weather corruptions. All are publicly available and will be used within the stipulated licencing agreement stated by the data owner.*

### 1.2 Types of data

*Datasets explained in section 1.2 consist of dashcam images of urban scenes. A team of researchers in Europe complied the images with stringent calibration techniques for trustworthy data. They intentionally collected the data when the weather was 'fair' meaning some realistic driving conditions are not represented. If more challenging scenes are needed Cityscapes has added additional data for synthetically applied fog and rain conditions with varying severity.*

### 1.3 Format and scale of the data

*Cityscapes contains data from 50 cities in Germany (and some neighbouring countries) and images are annotated with 30 semantic classes. The dataset contains over 20,000 coarsely annotated images and ≈5000 finely annotated images providing pixel-level, instance-level and panoptic semantic ground truth labels. Raw images and segmentation masks are provided in portable network graphics (.png) format. Data contains both 16-bit High-Dynamic Range images and 8-bit Low-Dynamic Range format to use at an image size of 1024 x 2048. The dataset is 11.5GB, since it is already a benchmark suite the FAIR principles (Findability, Accessibility, Interoperability and Reusability) are readily satisfied.*

## 2. Data collection / generation

### 2.1 Methodologies for data collection / generation

*No new image data or annotations are intended to be gathered or produced. Data comprising of results of the experiments will however be tabulated clearly with links provided to relevant repositories housing reproducible code - project transparency is of the utmost importance. Segmentation images from proposed networks will be stored locally and distributed on request. The FAIR principles of data will be adhered to whenever necessary.*

### 2.2 Data quality and standards

*Quality of the training data used (cityscapes) and generated segmentation results can be assessed via the following five dimensions: plausibility, correctness, currency, concordance, and completeness. Stochasticity introduced via optimisation routines means repetition of network training may produce more robust results; however, depending on the time taken to train said models this may be impractical within the time constraints of the project.*

## 3. Data management, documentation, and curation

### 3.1 Managing, storing and curating data.

*Cityscapes data has been downloaded to a local machine and uploaded to a Google Drive account to be used in conjunction with the Google Colab Pro environment to leverage higher compute power. Training data backup is not considered an issue due to the public availability. Trained networks however, including architectures and weights, will be stored in training logs as .pth files which will be backed-up locally along with code in the project GitHub repository.*

### 3.2 Metadata standards and data documentation

*Descriptions of the network architectures and any hyperparameters used for training the models will be stored and supplied within the appendices of the main report. Any hyperparameter values needed for reproducibility will be specifically stated within a README created for the project GitHub repository, along with any training schema and algorithms. Time taken, machine specifications and environment will also be noted for each training and validation cycle. This also helps with benchmarking the efficiency of any models created where execution time is a priority in segmenting live images for autonomous driving applications.*

## 4. Data security and confidentiality of potentially disclosive information

### 4.1 Main risks to data security

*Since training data is entirely public data, security issues with third party storage solutions (raised by ncl.ac.uk [https://www.ncl.ac.uk/library/academics-and-researchers/research/rdm/working/](https://www.ncl.ac.uk/library/academics-and-researchers/research/rdm/working/) ) are not a concern. Indeed, even any trained networks we create push towards a philanthropic cause meaning public distribution is encouraged.*

## 5. Data sharing and access

### 5.1 Suitability for sharing

*Yes: The data I will be using is entirely publicly available and already highly peer-reviewed and cited in many published papers. This project aims to remove bias from dataset that exist in the wild, thus sharing code and experiments will be encouraged.*

### 5.2 Discovery by potential users of the research data

*All code libraries will exist within the public project GitHub repository ([https://github.com/JackStelling/BiasMitigation](https://github.com/JackStelling/BiasMitigation)). If the project successfully manages to fulfil its aim, publication will be sought due to the universal application across domains for fair AI, whence a DOI will be generated to facilitate discovery.*

### 5.3 Data preservation strategy and standards

*The code to reproduce experiments will be hosted within GitHub indefinitely, but this requires the availability of the cityscapes dataset to reproduce exact results – this is not the authors responsibility. Training schema can be applied to more current datasets in future if desired. Evaluation data from the experiments will be shared; however due to the size of the image data, segmentation image results will not be hosted in a repository – detailed steps of how to reproduce these results will be provided.*

### 5.4 Restrictions or delays to sharing, with planned actions to limit such restrictions

*Proposed procedures for data sharing will be explicitly stated in the repository README, due to the fully public nature of the datasets used, confidentiality is not an issue. Intellectual property arising from developed models are to be freely distributed for advancements of the field; terms will be explicitly stated within the project README. The author will not assume any liability for misuse of any developed software.*

| 6. Responsibilities and Resources |
|---|
| *Are there any resources (e.g. storage/ training) that you will require to fulfil the plan?*<br>*Google Colab Pro @ ≈ £8 / month*<br>*Google Drive Storage (100GB) @ ≈ £1.50 / month* |

| **7. Relevant institutional, departmental or study policies on data sharing and data security** | |
|---|---|
| **Policy** | **URL or Reference** |
| Data Management Policy & Procedures | https://www.ncl.ac.uk/media/wwwnclacuk/research/files/ResearchDataManagementPolicy.pdf |
| Information Security | *https://services.ncl.ac.uk/itservice/policies/InformationSecurityPolicy-v2_1.pdf* |
| Other | |